

THE
MUSEUM GAZETTE.

No. 2,

JUNE, 1906.

VOL. I.

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*** START OF THE PROJECT GUTENBERG EBOOK THE HASLEMERE
MUSEUM GAZETTE, VOL. 1, NO. 2, JUNE 1906 ***



The Historical Department.

THE MUSEUM GAZETTE.

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VOL. 1.

OUR HISTORY ROOM. (*See Frontispiece.*)

A department of our Museum to which, as regards its educational usefulness, we attach very great importance, is that which attempts the illustration of Human History. It is displayed in a separate division of the main building, and is arranged, as far as possible, on “the space-for-time method.” This method, which, following the pattern of an ordinary diary, allots to every period of time the same amount of space, is, of course, possible only where the time-periods and dates are fairly well established. It is not well adapted, excepting as a sort of open and, to some extent speculative, framework for the illustration of prehistoric times. A courageous example of such use of it we ventured to offer in our last number in reference to prehistoric man in Britain. It was not history in any other sense than that the periods of time were real; the events assigned to them were largely conjectural. In the Museum itself we do not attempt to deal with very remote periods in this manner. Our space-for-time arrangement begins only with 2000 B.C. It might now, perhaps, fairly begin with 4000 B.C., but, unfortunately, we have not space enough. In this Schedule, which occupies the whole of one side of a long room (70 feet), a measured space on the wall, of nearly two feet, is allotted to each century. The centuries are marked out by strong black lines, drawn vertically from roof to the table-shelf below. This table-shelf is 18 inches wide, and runs the whole length of the room. It is upon it that the busts shown in our frontispiece are standing. Each bust is supposed to be in its appropriate century, and with it are placed any other illustrative objects belonging to the period—medals, coins, small architectural models (when we have them), and the like. For instance, a model of Stonehenge stands in the century in which it seems probable that that most remarkable structure was built, and portions of Roman pavement and other relics mark the period of the Italian occupation of Britain. Upon the wall itself are placed engravings, photographs, and the like, illustrative of the century, and representing either human personality or some results of human effort. In order to aid the memory each century is designated by the name of some prominent person of the time, to whom other associations may conveniently cling. These names, painted in bold characters, head the columns which represent the centuries. Beneath these prominent names we have (in the case of a considerable number of the most recent centuries) put up schedules of the principal events, and lists of some of

the principal persons. The appended schedule is one of them and will illustrate what is meant:—

FOURTEENTH A.D.

CHAUCE.

The Three Edwards.

Bannockburn.

Famine in England.

The Hundred Years' War begins.

Battle of Crecy.

The Black Death (Plague).

Battle of Poitiers.

Bolingbroke dethrones Richard II.

Froissart's Chronicle.

Wallace and Bruce. Dante. John of Gaunt. Rienzi. Van
Artevelde.

Wickliffe. Huss. Boccaccio. Petrarch. William Tell.

It will, if what we have tried to describe has conveyed its intended meaning, be seen that an observer passing slowly down the length of the room, may appreciate at a glance the relative position of the principal events in the world's history. He can hardly avoid noticing, with fair accuracy, the distance between Homer and Socrates, between Socrates and Paul, and between the Christian epoch and the times of Milton and Shakespeare. He will be impressed at once, as, possibly, he never was before, with a perception of the brief and very recent portion of time which contains the whole of the annals of our own nation. If, in addition to thus obtaining a sort of bird's-eye view of the progress of the world, it is desired to go into detail and devote time to the enquiry, a certain amount of help will be found to have been provided on the table-shelf. Detailed schedules taken from the "The Centuries" (see advertisement) have been mounted on board conveniently for hand use, and are placed on the table-shelf at the foot of each century. A few books of reference in biography and history, and numerous maps, have also been suitably placed, and there are chairs.

The "Historical Schedule" described takes up, as we have said, the whole of one side of the long room. The other side of it, as well as much of the floor-

space, is occupied by somewhat miscellaneous illustrations of prehistoric times, and of nations and races which have not as yet attained to history. The anthropoids, anthropology and ethnology in general here find illustration, in large part, but not wholly, by pictorial aid. We have also a few interesting objects suitable, as illustrating social progress, for what is now known as a Folk-Museum.

It is believed that this department of the Museum offers special facilities to teachers, who bring their classes into it and give explanations on the spot, and that by enabling the pupil to obtain a wide purview of historical times, it may do somewhat to obviate the inevitably cramping influence of the too detailed study of single epochs.

FAMOUS WOMEN AT THE NATIONAL PORTRAIT GALLERY.

We have climbed to the third landing. Let us turn into the left-hand gallery and we shall come to one of the most interesting groups in the whole Museum.

It is that which contains portraits of English women whose names have become famous in literature. Here we have Mrs. Browning, Sarah Austin, Mrs. Carter, Miss Strickland, George Eliot, Mary Somerville, and many others. The collection is not nearly so complete as could be wished—for Jane Austen, Anne and Jane Taylor, Hannah More, the Brontës, and many others are wanting—still, it is very good. But few males are admitted. Robert Browning is very properly allowed to accompany his wife, and the fact, we presume, that his wife was with him, has also gained access for Thomas Hood. A portrait of Lady Hamilton strikes us as a little out of place, but the Museum has as yet no department for female charm, and as this is by Romney it may have been difficult to refuse it. With the exception of it and one of Elizabeth Fry, all the others have more or less direct claim to be associated with literature or science. To Mrs. Opie, Mrs. Browning and Miss Strickland no one will hesitate to accord the praise of good looks, and many others exhibit in a remarkable degree the bright-eyed intelligence which we expect from authoresses. Several show a splendid breadth of forehead, the accompaniment, no doubt, of a brachycephalic or broad head. Mrs. Carter, Miss Mitford (of “Our village”), Mrs. Trimmer, Miss Strickland and George Eliot are the best, but not the only instances of this. It might be hardly good manners to describe too exactly the various features of feminine faces, and the fact that but few show the profile makes it difficult to judge accurately as to size and shape of nose and chin. Few are, in any sense, disappointing. That of Mrs. Carlisle might have been kept back without loss, since, if it is in the least true to life, it gives a too painful sense of justification to the rumours of married unhappiness, which were probably to a large extent unfounded. The portrait of Sarah Austin, when old and ill, might also perhaps be spared, since there is a very pleasing one of her in earlier life. At any rate, the two ought to be placed together. Declining, as we do, in reference to almost the whole, the task of detailed analysis of features, we cannot, in the interests of physiognomical research, exempt those behind which lay the most profound intellect ever possessed by a woman. Mrs. Carter in classical and literary

attainments, and Mary Somerville in the domain of science, must be accorded foremost places. Mrs. Browning is second, perhaps, to none in depth of human sympathy and beauty of poetic expression; but if we estimate character by profundity of insight, we shall probably accord to George Eliot amongst women much the same position as that which Shakespeare holds amongst men. We do not for a moment compare her with Shakespeare.

Of George Eliot the Museum possesses three portraits. Most fortunately, it has also one of her father. It would add enormously to the value of portraits as a means to the illustration of character, if we might always have associated with that of a distinguished individual those of his parents, and even of his brothers and sisters. The portrait of Robert Evans (George Eliot's father) is a very pleasing one—a grave, serious face, with a large Roman nose, well-formed lips and chin, and a really magnificent forehead. The nose probably gives a clue to his family descent.

Of those of the authoress herself, the first, taken when she was 23, by a lady friend (Mrs. Bray), is a poor work of art, and exhibits a commonplace face, surmounted by a very large rounded forehead. It is impossible to judge of the nose. A second is of some years older, and is much better executed. The forehead is still there, and the nose is shown of good size and shapely, and the lips and chin are well formed. The face is a shorter one than in the next, and the hair is of a much lighter tint. The face is pleasing and attractive, not much unlike, if we remember rightly, one which represented Jenny Lind.

Lastly, we come to the often copied and well-known portrait at age 46, by Sir F. Burton. In this the nose and face are long, almost suggesting an approach to what is called a horse face. The chin is good, but as the fine forehead is much concealed by folds of hair, the balance of features is not perfect, and the whole result not pleasing. All the three show the eyes light grey or blue. There can be no doubt that George Eliot had a large forehead and a fairly large nose and chin, but further than this these somewhat disaccording portraits do not take us. We must fall back on her father's Roman nose and really beautifully balanced features.

Mary Somerville's face is a very intellectual one, but a trifle cold as regards mouth and chin. Looking at Mrs. Carter's width of forehead, no one can doubt that she had "skull-room" for many languages and what had been written in them. There is a most pleasant expression on her face, suggesting a genial companion and ready conversation, but her chin is weak and small. Mrs.

Browning's face is alive with graceful vigour, and her forehead, as we have already said, is full and round.

It is held that the heads of women are dolichocephalic (long in proportion to breadth) in larger average than those of men. There is certainly nothing in this collection of portraits which would oppose the supposition that the higher developments of intellect in women, as in men, are usually met with in heads unusually broad. Such an inference must, however, be accepted with some caution.

THE BRAIN IN RELATION TO INTELLECT.

When we speak of size of brain in reference to intellectual endowments we must draw a clear distinction between the eminence of a specialist and that of one of wide attainments. A man may become famous as a specialist by the sedulous cultivation of one branch of knowledge, whilst far from being distinguished by wide grasp of thought. Indeed, the absence of interest in some branches of knowledge may greatly favour the exclusive devotion to a single one. Peschell quotes the weights of the brains of some Göttingen professors which were considerably below the average: Gauss, Fuchs (pathologist), Hermann (philologist), Haussmann (mineralogist).

Cuvier and Lord Byron are usually cited as having had brains much heavier than the average, but there is some uncertainty, especially as to the latter. None of his lordship's portraits suggest a large head.

Although there are considerable discrepancies in the statements of those who have examined the matter, there can be little or no doubt that the brain increases somewhat with the advance of civilisation, and that it was smaller in the prehistoric races of man than in the more advanced ones of the present age. The differences are probably not so great as many may expect, but they are real. The Australian natives stand the lowest, and the Europeans the highest. The American Indian had a larger brain than the Asiatic, and the Asiatic than the African. The Chinese stand between the European and the Negro. Two Irish skulls are perhaps the largest on record.

The size of the skull may possibly not be always a safe indication of the amount of useful brain matter contained in it. It is said that the large-skulled Germans have brains which are of lower specific gravity than those of others.

There are many sources of fallacy as regards the weight of the brain in different persons, different races, and in the two sexes. We must not trust too implicitly to statistics or to tabulated records. It is better to be content with general results, and to state even these with great caution.

There can, however, be but little doubt that the brain of woman weighs less than that of man, and that this difference is greatest in highly civilised races. The brains of the broad-headed are, as a rule, somewhat heavier than those of the long-headed. This conclusion has been arrived at by comparing the brains of

different individuals of the same race, not those of different races (Peschell, p. 70).

There are some observations which support the conclusion that the brain attains its greatest weight before 30 years of age, and then undergoes diminution. At the age of 80 this diminution is supposed to reach 10 per cent. The diminution concerns the brain proper and the cerebellum, but not their connecting part, the pons, which increases up to the fiftieth year. Whilst it is, however, difficult to imagine methods by which, without risk of great fallacy, such conclusions could be arrived at, we may safely believe that the advance of age is attended by some reduction in the size of the skull cavity and the weight of its contents.

It appears to be a constant law that with advance of civilisation the differences between the sexes in general become increased. This is seen in measurements of the skull and in the weight of the brain. In the brains of negroes the woman is but little below that of the man, 984 to 1,000, but in the English it is only 860, and Germans 838. These figures indicate, of course, proportions only, and like all other calculations in this difficult subject, must be received with caution; but they probably indicate, if they do not precisely express, the fact.

The height of the skull is usually in inverse ratio to its breadth. The variation in height is far less than is common in breadth.

The brachycephalic have heavier brains than the dolichocephalic.

In Hottentots both indices, breadth and height, are low.

The following may be mentioned as examples of remarkably broad heads (wide foreheads), in association with genius: Shakespeare, Beranger, Mirabeau, Peacock (the novelist), Miss Austen, Blackmore ("Lorna Doone"), Tennyson, Erskine.

The following had heads both tall and broad: Scott, Goethe, Cervantes, Ambrose Paré, John Foster, Father Paul, Galileo, Michael Angelo, Machiavelli, Benjamin West.

We shall be indebted to any of our readers who may direct our attention to other well-marked examples.

The following afford instances of remarkably tall heads: Motley (the historian of the Netherlands), Remin (engineer), Richard Roberts (engineer).

The following had remarkably long faces: Francis I., Inigo Jones, George Eliot (Miss Evans), Savonorola.

THE EGG MARKET IN ENGLAND.

A wholesale dealer gives us the following items as to the consumption of eggs in London: When eggs are 16 for a shilling the sale amongst the working classes is enormous. When they are 12 a shilling it falls at once 40 per cent., and when only 8 it drops very low. The bulk of London eggs are imported; Italy supplies many, France many, and some even come from Turkey. They will travel from Trebizond and be perfectly fresh when they arrive in London. We are assured that most of the new-laid eggs consumed in Haslemere come from Italy, and this at all periods of the year. Our English housewives are not clever at preserving eggs, and the witty classification of eggs into new-laid eggs, fresh eggs, and eggs, is but too often illustrated.

ON OUTGROWTHS AND APPENDAGES.

(Part of a Museum Lecture.)

Let us draw a clear distinction between “Appendages” and “Outgrowths.” The prickles which are formed on the branch of a rose are appendages; they may be detached without really breaking any part of the bush. The spines which grow on the blackthorn are outgrowths, and cannot be so detached. I wish that some better word than “appendages” could be found, for it seems almost to imply insignificance, and many of the appendages to plants are of the utmost importance. Still, it is true of them all that they may be removed and yet leave the plant, as a plant, complete, and many or most of them have only a transitory life, which does not by any means equal that of the plant itself. They are like the luncheon basket at the summer day’s ramble, not absolutely essential, but very conducive to perfection. Now most forms of leaf, flower, fruit and prickle are in this sense merely appendages. From the very earliest stages of their formation arrangements exist for their separation, in whole or part, from the plants on which they are produced. You will see that I am cautious in my terms, and say in part or in whole, for in truth some appendages never are detached as wholes, and very great variety exists in the ways in which they are dealt with. For the most part they are susceptible of death, and have their fixed duration of life quite independently of the plant which bears them. In many this independent death is the cause of their being cast off. In some instances, however, it is not death, nor even sickness, but the fulness of life and the attainment of adult age which causes them to leave the parental home. I hold in my hand an oak twig with two empty acorn cups. The acorns having attained maturity, have fallen out. Shall we say they have detached themselves, or that the tree has detached them? They have not fallen by mere weight, for they were doubtless nearly, if not quite, as heavy whilst still green, and they were then firmly fixed. You see at the bottom of the empty cup the large round scar which marks the site of former attachment. It is brown and dry. It was by changes which took place here that the acorn was loosened. The acorn had ripened and ceased its growth. It no longer attracted sap through its base of attachment, and the latter consequently became dry and brittle. Possibly its feeding tubes were choked; at any rate, it is certain that it underwent a sort of death and was no longer able to keep the acorn in place. The process was much like that which occurs in the shedding of

leaves, with, however, the very noteworthy difference that the acorn itself was still alive.

We have not, however, done with our oak twig. The acorns which it bore were only appendages to an appendage, and it now becomes the turn of the cups themselves and the whole of the long foot-stalk on which they are mounted to become detached. These are no part of the tree, and are of no use to it. They were developed in order to bear flowers and fruit; that function they have now discharged, and they must die. Life is preserved only by the discharge of function, or at any rate the effort to discharge it. Utter inactivity leads to death, and death leads to separation from the living and to decay. You see that the whole foot-stalk is brown and shrunken and evidently dead. This condition ends abruptly where the foot-stalk joins the stem. At this spot, if you look carefully, you will see that there is a ring of constriction, marking definitely where detachment is in progress. This was the spot at which the production of the whole appendage began, and here a sort of joint was left at which the final detachment was destined to occur. Just one word of caution, that we must not carry our distinctions too far. After all, they are to some extent matters of degree. The joint which separates the appendage from the twig on which it is produced can hardly be termed a true joint, for certain structures run in unbroken continuity from the stem to the appendage. These are the fibro-vascular bundles by which the appendage is fed and also fixed in place. These bundles are usually quite visible in the scar-surface left when a leaf or fruit is broken off. They are “the nails in the horse-shoe” of the leaf-scar of the horse chestnut. Still, it is certain that a sort of joint is present, and that the structures are continuous in a very different sense from that of a stem or true branch. You may observe this difference in my acorn-bearing twig, for there are two acorn cups, and one has been produced by a branching out from the stem of the other. This little branch is smoothly continuous with the parent branch, and shows no preparation for detachment whatever.

Thus we have seen that the arrangements under which leaves are shed are exactly repeated in the case of fruits, and that it is by no means needful that the object to be detached should be dead or dying. It may perhaps surprise you to be told that sometimes appendages are shed which have by no means accomplished their prospective work. Some plants shed their flowers and do this deliberately, having made their arrangements for a step which is apparently suicidal. In reality it is not suicidal, nor is it one of limitation of population, but simply of preferential employment of capital. The potato gives a good example of this. Every spring you may see on the heads of this plant beautiful flowers produced,

which are destined in the course of another week to be only flowerless foot-stalks. The flowers break off at a pre-existing joint, just as leaves are shed. The influence which causes them to fall is inability to attract sap, in consequence of inability to proceed to the further stage of producing fruit. The young tubers underground make such overpowering demands upon the sap-furnishing capabilities of the roots that the flowers cannot obtain sufficient for their seed forming. Thus they at once die: if not obviously, at any rate practically, and detachment follows as a natural result.

It is a case of competitive growth and the tubers win. After a time the plant will in the course of inheritance learn that it is useless to produce flowers, will give up the attempt; indeed, many varieties have already done so to a considerable extent.

No better instance could perhaps be given of the law which goes through all animated Nature that activity is almost essential to continuance of life, whether in individuals or their parts.

THE OLDEST FOSSILS (Lingula).—It is a noteworthy fact that these, the oldest fossil animals known, belong to species by no means low in the scale. Yet every trace of the many millions which must have preceded them, and have gradually led up to their development, have perished. The period of time which must have elapsed subsequent to the advent of life upon the planet and the development of the lingula mollusc was probably quite as long as that which has passed since the lingula left its shell in the mud of the Portmadoc slate. The oldest fossils which are known are found in the lower Cambrian rocks. They are small oval shells, which were, during life, the protection of small soft-bodied sea animals of highly complex structure: they had red blood. Their descendants are still found in great numbers burrowing in sand on the shores of tropical oceans. They have received the name of Lingula, and have in turn conferred that name on certain hard rocks in which their shells occur in abundance, the “Lingula flags” of Wales (RAY LANKESTER).

HUGE SHARK’S TEETH.—Ray Lankester figures, in his interesting lectures on extinct animals from which we have quoted the above a gigantic shark’s tooth.

It is that of the *Carharodon megalodon*, and is three times the length of the tooth of any living shark. Specimens of this fossil tooth of smaller dimensions are common, and one should be found in every museum. They are obtained from the bone bed of the Red Crag at Felixstowe, but were not originally deposited in it. Many of them have fragments of a yet older sandstone adhering to them. Lankester calculates that his shark was 100 feet long.

THE GREAT FAMILY OF THE CAMELS AND DEER.

The Camel, the Camelopard, the Musk-deer and the Deer are all more or less nearly related. They constitute a branch of the great family of Ruminants, and all chew the cud. With the Camel are associated the Bactrian or two-humped Camel, the Alpaca, the Llama and the Vicuna. With the Giraffe we have the Okapi, and several extinct animals. The Musk-deer stands almost alone. Of Deer there are a great many species. It is easy enough to distinguish these various animals the one from the other when seen living in a Zoological Garden or stuffed in a museum. Indeed, at first sight, there might seem to be no great similarity between a Camel, a Giraffe, and a Fallow-deer. To the student of natural history, however, it becomes of great interest to observe the essential peculiarities of each. These may be grouped as those which prove relationship and those which show differences. We will leave aside the very important peculiarities in the stomach, because but few of our readers will have opportunities for examining them, and will confine our attention to the feet, horns, skull and teeth. All have two hoofs or more, and the Camel group have behind their hoofs a pad which covers the sole. None of them have hollow horns, and in none are their weapons of offence—horns, teeth, tusks, &c.—very effective. In all when adult the cutting teeth (incisors) in the upper jaw are absent, and in most the canines are either absent or much modified.

The Camel tribe differ from Giraffes in possessing a pad, and in having, when young, incisor teeth in the upper jaw, and fewer lower incisors by two. They have also strong canines in both jaws, no trace of horns, and nothing to be called tusks.

The Giraffes have two, three, or even five abortive horns of very peculiar development. They have very long necks, no pads, no tusks. The canine teeth in the lower jaw, which look like incisors, are much specialised in being cleft or notched.

The little Musk-deer has no sort of horn; but his upper canine teeth are large and form tusks.

The true Deer have antlers (in the male), which they shed every year. The males, and sometimes the females, have canine teeth in the upper jaw. The

antlers are dermal bones, that is, are formed in the skin, and do not grow from the skull. They have two rudimentary digits above the hoofs.

The whole of this group, which we may call the Camel and Deer family, are almost wholly defenceless, the Giraffe the most so of all, and, excepting those which are useful in domestication, are threatened with extinction. The Camels and the Llamas, although separated as distantly as Peru and Arabia, have in common the very peculiar habit of snorting most offensively at those who oppress or annoy them.

This large family of Camels and Deer stands between a small one which comprises Pigs, and a very large one, to which Cattle, Sheep, Goats and Antelopes are assigned. Pigs are not ruminants, and have incisor teeth in the upper jaws. Cattle, sheep, &c., like camels and deer, ruminate, and have no cutting teeth in the upper jaw. Their distinctive features are hollow horns (which are present in both sexes), and the invariable absence of tusks.

The canine teeth in all members of the Giraffe group are peculiar, in that they show a cleft in the free edge which divides them into two lobes. These teeth look as if they belonged to the incisors, but various facts prove them to be really the canines. Those of the extinct *Sivatherium*, and those of the recently discovered Okapi, have similar peculiarities, and thus prove their relationship.

It is curious that our natural history authorities are not yet agreed as to whether the Giraffe has his fore limbs longer than the hind ones or not. Claus and Sedgwick say, "hind legs much shorter, and therefore the back slopes backwards." Those who have measured the bones, however, say that there is no difference, and that the slope depends entirely upon the setting of the shoulder blade.

ENGLISH EDIBLE SNAILS.

“WALL-FISH.”

The term “wall-fish” will be unknown to many of our readers. It is applied by dealers in Covent Garden and other markets to the common garden snail (*Helix aspersa*). This mollusc is held in especial esteem by the poor in Bristol, and in consequence is now very scarce in the environs of that city. There are men who make a livelihood during the winter by collecting these snails from their hibernating places. In November, 1896, the writer met a “wall-fish” collector in a remote village in Somerset, and had an interesting conversation with him. He was collecting for a Bristol dealer, his home being in Kent, where he worked as a carpenter in summer and autumn. For many winters past he had regularly visited Somerset to collect snails. According to his experience these snails seldom hibernate in banks facing east or north, but usually seek winter quarters in those facing south-west.

They generally congregate in some numbers, and appear to have a predilection for certain spots. From an hybernaculum near the village he had that morning taken a gallon and a half of them, but this was very unusual; his “takings” as a rule did not exceed a gallon per day.

Ash stumps, or crannies at the base of ash-trees, are very favourite haunts. They seldom hibernate under oaks, and although old walls are favourite places in summer, whence they probably owe their name of “wall-fish,” they seldom hibernate in them.

Our “wall fisherman” carried an iron rod about 2 feet long, slightly crooked at one end for probing likely nooks and corners. *H. aspersa* is quite the most nearly domesticated of the snail tribe. It loves the haunts of man, and is seldom found in any numbers in places remote from villages and roadsides.

A conchologist should never miss an opportunity of examining the bag of a wall-fish collector. Occasionally rare varieties—*scalariforme* for instance—may be secured in this way. I asked my friend to allow me to inspect his “catch,” and he very obligingly turned out the contents of his creel. I found nothing better than three or four examples of the variety *exalbida*, which is greenish-white. It is, however, widely distributed in the southern counties. It is stated by Forbes and Hanley (“Hist. Brit. Moll.,” vol. iv., p. 46), that “owing to its being an

article of food in some countries, or else a supposed remedy for pulmonary affections, *H. aspersa* has been transported and distributed by the agency of man to all parts of the world. It is especially abundant in the neighbourhood of gardens." In 1840, according to Turton, *H. aspersa* was sold in Covent Garden and elsewhere as a cure for diseases of the chest, and was sent to the United States as a delicacy. "The glassmen at Newcastle once a year have a snail feast; they generally collect the snails themselves in the fields and hedges the Sunday before the feast day." This feast is, we believe, now given up.

DORMANT LIFE.—The conditions under which vital activity may become dormant obtained a curious illustration in the instance of a beetle which was taken alive out of the wood of a desk which had been in the office of the London Guildhall for twenty years. The wood was deal from the Baltic, the beetle was the *Buprestis splendens*. It was alive and in beautiful colour. The observation was confirmed by Sir Joseph Banks. The description is given by Mr. Thos. Wrenham in the tenth volume of *Transactions of Linnæan Society*, 1810.

DODDER AND IVY.—Ivy is a climber only, Dodder is a true parasite. Ivy does not in any degree derive nourishment from the trees on which it grows. It is obvious that it cannot get any from walls. It may be plausibly disputed whether it does any injury to the trees to which it clings, for it is often seen on very large ones. It is reputed to be wholesome for sheep and deer in spite of its rank odour, and pheasants are fond of its seed.

HOW TO FORM A TEMPORARY MUSEUM.

A temporary museum will, in all probability, be a Vacation- or Summer-museum; there will, therefore, be no need for stoves or fires, and scarcely any for artificial light. The sun rises in summer as early as any members of the museum committee are likely to be stirring, and by sunset it will be time to close. We will suppose, then, that the season is summer and the place a small town in the country. In the first place a small local committee of those interested in the scheme should be formed, and a small sum of money guaranteed. The next step should be to borrow a set of school premises, or hire for a couple of months an empty house. If only a small cottage were obtainable it should have a back garden in which a large wooden shed could be put up. The essentials are plenty of room, plenty of light, and good protection from weather. If the premises secured be those of a school, the next thing will be to arrange with some carpenter for the hire of a quantity of boards which, laid across the desks, will make tables. It may be possible, on similar terms, to obtain from a draper a quantity of baize, or its very cheapest equivalent, but this would not be absolutely necessary. Having secured plenty of table-space, the next point is to prepare the walls. It will be required to display on these, Portraits, Maps, Illustrations, &c., &c., and for pinning these up some sort of framework is desirable. The carpenter will soon put this together and cover it with baize or flannel.

The rooms being made ready, the next step is to fill them, and about this there will be no difficulty. As a preliminary measure a circular will have been sent out, inviting all residents to contribute their curiosities on loan. It will be strange if this be not bountifully responded to by cases of stuffed animals and birds, collections of eggs and of shells, and boxes of minerals. Ammonites, elephants' teeth, mammalian skulls, butterflies, wasps' nests, flint implements, and Missionaries' curios will be brought in great abundance. All these, properly arranged, with descriptive labels, may be made most interesting and instructive. A certain number of glass-covered display-cases, with locks, will now be required in order that fragile or valuable specimens, and especially those on loan, may be properly taken care of. Such cases may be made inexpensively, and our Haslemere Museum will be glad to supply patterns, or even, if wished, to loan the cases themselves.

At this stage the Committee should remember that a museum has been well defined as “a collection of labels illustrated by specimens,” and should obtain, if it has not already done so, a set of our printed labels, and ascertain for how many of them illustrative specimens can be produced. Steps should next be taken to obtain elsewhere any specially desirable exhibits which may not be forthcoming. Some Horns, Antlers, Skulls, &c., are sure to be wanted, and no doubt the Geological series and the Flint implements will need to be supplemented. These deficiencies may be supplied in some instances by borrowing from other adjacent museums, or they may be purchased at various dealers.

A very important and attractive department of the temporary museum will be the display of Pictures, Maps and Portraits. These, like those just mentioned, may be hired, if they cannot be begged or borrowed. A diligent ladies’ sub-committee, well supplied with bundles of old unbound copies of *The Illustrated London News*, *Graphic*, *Vanity Fair* and *Punch* would soon construct an attractive portrait gallery, as well as sundry most interesting series in illustration of social history, scenery, geography and natural science. There is not anywhere a small town in which material of this kind may not be brought to light from the cupboards in which it is uselessly stowed away. The best way of dealing with it would be to procure some false-backed frames, such as we have in use at Haslemere. In many instances portraits, maps, &c., will not need to be put into frames, but may be at once pinned up in well-classified series on the walls.

We have said nothing as yet as to a department which ought to be made one of the most important, especially at a sea-side resort. A Vivarium for the display of local specimens in their fresh and living states should be arranged either in an ante-room or hall, or in a shed, or under a verandah outside the building. In this should be a stand for flowers (all named), and bell glasses and large saucers for the reception of shell-fish, sea anemones, sponges *au naturel*, sea-weeds and corallines. Illustrations and explanatory labels for most of these our Haslemere press can supply.

The charges for admission to the show should be: before its completion, sixpence each person; when complete and in good order, threepence for adults, half-price for children; and on Saturdays a penny all round. Books of tickets, making a very liberal reduction, should be available. The result would be, if circumstances were favourable and zeal abounded, that the guarantors would lose nothing and might possibly carry forward a modest balance to begin next year with. Meanwhile the prosperity and reputation of the sea-side resort would

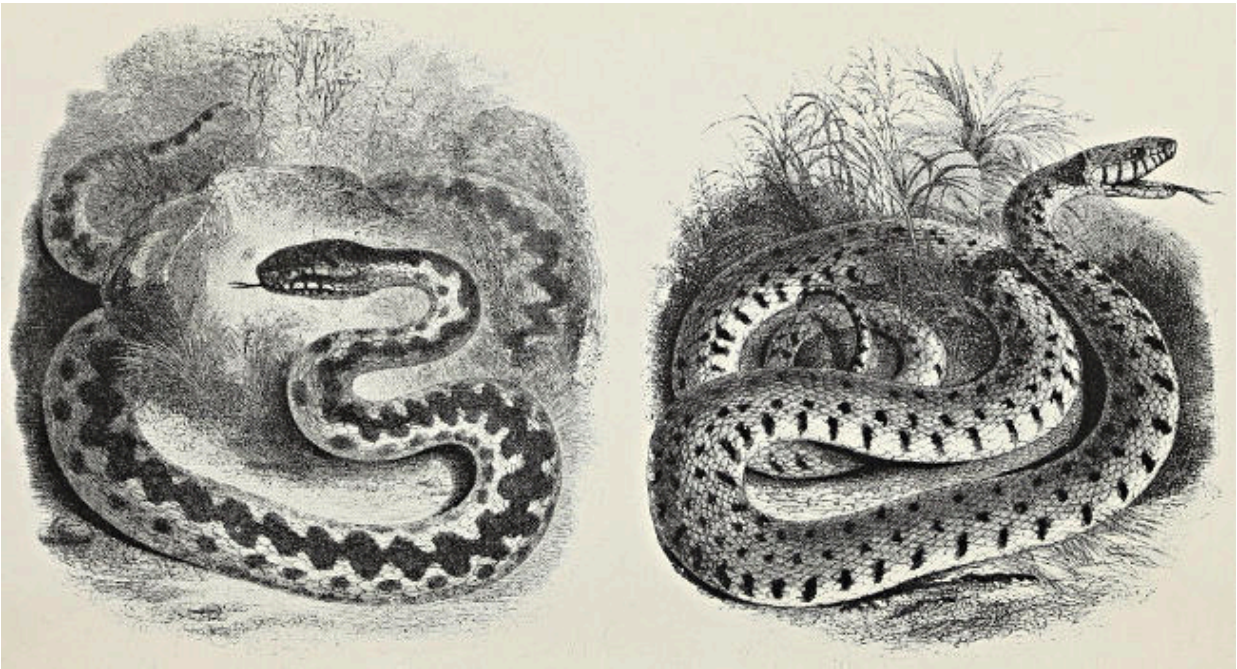
have been helped and a large number of persons would have been entertained and instructed.

The following paragraph appeared a short time ago in one of the daily papers:

The complaint is often made that there is little opportunity to learn anything about the specimens exhibited in museums. At Brooklyn the experiment has been tried of placing for the use of visitors books bearing upon the subjects exhibited alongside the cases. This experiment seems to have been most successful.

The Brooklyn experiment is not a new one. It has been practised for many years past in our museum (*vide Museums' Journal*, vol. ii., 1902).

BRITISH SNAKES.



The Viper. The Common Snake.

We have in Great Britain only three representatives of the class Reptilia which come under the name of Snake. They are the Common Snake, the Smooth Snake and the Viper. In Ireland there are—as the result of events to which we referred in our previous number—none at all. Of the three British Snakes, one, the Smooth Snake,^[1] is so rare that it is of interest only to the naturalist. It is met with in Hampshire and Dorset, and perhaps in Scotland, but it is nowhere frequent. It is more nearly allied to the Common Snake than to the Viper, and is quite harmless. It is much smaller than the others. We will concern ourselves for the present only with the other two. The Common Snake^[2] is quite harmless and should never be injured by any humane person. The Viper^[3] is venomous, and should be destroyed without mercy. It is easy enough to distinguish between the two even when in movement. The Common Snake is always, when full grown, much longer than the Viper. It may measure 4 feet and is usually 3, whilst the Viper is never more than 2. The Common Snake tapers off very gradually at its tail, whilst the Viper has a short tail, which is abruptly constricted at its base. The Viper is usually brown, often deeply coloured, whilst the Snake is much lighter coloured, of a light grey-brown tinged with green.

The Common Snake has only spots of black, whilst the Viper is marked down the whole of its back with large zig-zag black lozenges. This is a most important feature, is characteristic at all ages of the animal, and is easily seen under all conditions.

We have named the characters which are most easily seen when the animal is gliding about on the sward or path, there are others which are available when the animal is dead or in captivity. Under such conditions it will be seen that the scales which clothe the Viper are much smaller in size than those of the Snake, more especially on the head, and further, that the head of the Viper is marked with black, somewhat in the form of the letter V.

Snakes, as well as Vipers, have sharp teeth, but they use them only for seizing their prey, and they have no poison fangs.

The venom apparatus of the Viper consists of a gland which secretes the poison, a receptacle which stores it, and a long, sharp fang, which can be extruded and through a groove in which the poison is conveyed. The animal darts open-mouthed at its enemy rather than bites. Its fang being in its upper jaw it becomes well exposed when the mouth is agape. As there is a fang on each side there will usually be two punctures in the skin about a third of an inch apart.

No doubt there occur every year in England a good many instances of viper-bites in men and boys. There are, however, exceedingly few deaths, and many persons of considerable experience have doubted whether the bite is ever actually fatal. A very urgent and severe illness is almost invariably the result; but the patient just pulls through.

In a case in which the writer was, in boyhood, a particeps, two fine vipers were captured in mistake. They were put into a botanical box and were repeatedly inspected, and their tongues freely touched. It was only when on arrival at home that one of them, during an attempt to transfer it to a cage, made a dart and struck the finger of a boy of fourteen. The accident was concealed until, about a quarter of an hour later in attempting to cross the floor, the victim fell down in a deadly faint. He became very sick and having been got to bed remained in collapse, apparently near death for several hours. Brandy was of course freely given.

The medical adviser who had been summoned, arrived in hot haste with a big volume under his arm, in order to make reference to "snake-bite" and its treatment. It was, however, too late to do anything more than give stimulants,

and happily these were successful. A fortnight's illness ensued, during the early part of which the whole arm was enormously swollen, and later partly covered with boils.

This case affords, we believe, a very fair example of what usually follows the incautious capture of the English adder. The reptile does not attack willingly, but only after much provocation, and he cannot, as a rule, destroy the life of his human enemy. His poison fangs are designed for other purposes, and for much smaller animals.

Within the last few weeks a case at Folkestone has attracted much attention, in which two school-boys in pursuit of what is called "Nature Study," were bitten by a viper, with the result that one of them died. Several other well-authenticated instances of death have been recorded. We repeat, however, that they are very rare.

The early summer is the best time for killing vipers, since the males, which at other seasons hide themselves, may now often be found. Warm heaths and banks exposed to the sun are the places which they frequent. The common snake, on the other hand, loves water, or may be found in a hay-field or near an old barn or on a dung-heap. The snake lays eggs and leaves them to be hatched by the heat of the dung or the rays of the sun, taking no care whatever for her progeny. The viper, on the contrary, brings forth her young alive and tends them carefully, even allowing them, according to fairly well-accredited narratives, to retreat in case of danger into her mouth and gullet.

The first aid in case of viper-bite should consist in placing a tight ligature (string or a boot-lace) around the limb above the punctures. This should be as tight as possible so as to prevent the circulation of the blood and passage of the venom towards the heart. Next, incisions should be made across the punctures, or if circumstances are favourable, the bit of skin comprising the two punctures may be cut out. The wound thus made should be sucked or well bathed so as to favour bleeding, and to remove as much of the poison as can be got away. If ammonia or potash, or Jeye's fluid or Condyl's fluid be at hand the wound should be continuously bathed with a weak solution of it. The ligature, if tight, should not be kept on for more than an hour, but by this time medical advice will probably have been obtained. To combat the faintness, &c., brandy, or still better ammonia (that is, sal volatile or hartshorn well diluted), should be given.

[1] *Coronella lævis*, or *austriaca*.

[2] *Tropidonotus natrix*, or *Natrix vulgaris*.

[3] *Pelias berus*, or *Coluber verus*.

THE VIVARIUM. (HASLEMERE MUSEUM.)

During June and July the botany of a district may be very fully illustrated in the Museum-vivarium. Mr. Douglas Taylor, who has charge of that department in our Museum, experiences no difficulty in exhibiting one hundred species simultaneously. No very rare species are exhibited. At the present time the only plants in our collection to which the term “uncommon” may be applied are herb Paris, bird’s nest orchis, Solomon’s seal, and climbing corydalis. The flowers are arranged, in zinc cylinders, on an ordinary florist’s stand. For the printed labels now in use we are indebted to the generosity of E. E. Lowe, Esq., F.L.S., of the Plymouth Museum.

Vipers and grass snakes are not difficult to obtain (see page 53). The former, when discovered, should be carefully pinned with a stick, whilst a vasculum (or large bottle) containing leaves and heather is placed before it. Upon release the viper, judiciously guided by the stick, will take refuge in the receptacle prepared for it. Grass-snakes thrive well in captivity, their chief food being frogs and mice. On the other hand, the English viper nearly always refuses food under such conditions.

Our vivarium contains, in addition to the above, two of the three British newts, viz., *Lophinus punctatus* and *L. palmatus*. The former, though usually spoken of as the “common newt” is in many districts (as at Haslemere) not so common as the palmate newt. The latter is smaller, and the tail terminates abruptly in a threadlike filament, instead of gradually tapering to a point.

An ants’ nest contained between sheets of glass, on the plan devised by Sir John Lubbock (Lord Avebury) and described in his “Ants, Bees, and Wasps” (p. 2), always proves a source of attraction to visitors in the winter, as well as summer months. We have had for two seasons past a nest of the amber-coloured meadow ant (*Formica flava*) displayed under these conditions. It is probably the most intelligent of European species, forming the grassy hillocks from 9 to 18 inches high, so commonly seen in some districts. The queen, which is much larger than either workers or males, requires very careful searching for upon opening a nest; but the peculiar little white woodlice (which delights in the long name of *Platyarthous hoffmanseggii*, they are a sort of guest of the ants) may be always easily discerned.

Every spring we take out of the Museum and arrange in this department a series of our summer migrants, accompanied with general notes on bird migration. The following may be seen now: Swallow, martin, swift, sand martin, cuckoo, corncrake, nightingale, wryneck, nightjar, redstart, yellow wagtail, garden warbler, wheatear, chiffchaff and whinchat.

Kept under a bell glass and fed with lettuce and cabbage leaves are some half a dozen examples of the large edible or vine snail (*Helix pomatia*), the largest of our native land molluscs. This species occurs chiefly on the chalk in the southern counties. At one time it was supposed to have been introduced by the Romans, but of late years the opinion has gained ground that it was indigenous. In the same quarters is an example of a shell-slug, the *Testacella haliotidea*, sent from Torquay. The shell-slugs differ from ordinary slugs in having a shell on the tail. They are carnivorous and feed upon earth-worms, which they pursue in their burrows; hence may be considered as gardeners' friends.

A GILBERT WHITE PAGE.

(Continued from p. 27.)

The curious effect of insect attacks in stimulating the growth of certain parts of the plant which they have damaged may be found illustrated in hundreds of instances. We have just mentioned the pine-apples on the spruce fir, but the currant gall on the male catkin of the oak is yet more striking. In this case a structure which is naturally very shortlived has its vigour enhanced and its life prolonged by the presence of the parasite. In this instance the fly attacks the pollen-bearing flowers and deposits its eggs. These flowers would, in the ordinary course, wither and fall as soon as the pollen has ripened and been blown away. Under the stimulating influence of the larvæ, however, sap is attracted, their stems thicken and become fleshy, and instead of withering, they produce what looks like a handsome bunch of currants.

Another instructive instance of parasitism stimulating growth may be observed at this season in any patch of the common field thistle. Some of the plants are almost sure to be affected by a parasitic fungus. It is present in the stole of the plant, and its influence will cause the affected plants to put forth leaves earlier than the healthy ones. They will also grow faster, and in the course of a month be twice the height of their fellows. The fungus grows in the stem, and finally it will flower out on the surface of the leaves. When this happens the plant will die, but up to that period its growth has been notably vigorous. The fungus is the *Puccinia suaveoleus* (see Plowright, p. 183.) In some plants it is visible even in early spring, and if abundant will dwarf the plant so affected instead of stimulating its growth.

SEASONAL NOTES. JUNE.

It is scarcely too much to say that “leafy June” is the month in the year least favourable to natural history observation. Its glorious beauty is distracting and the profusion of objects of interest hinders attention to any. We hope, however, that our last month’s Notes may have directed the attention of some of our readers to certain special topics and particularly to the wonders of Gall-formation now in progress. The pine-apple galls on the spruce firs are now in full growth. Two varieties will be abundantly found. Some are small, not bigger than large peas, and remain green. These are covered with spines rather than scales. Others much larger have scales, which at their margins are beautifully tinged with various shades of red and crimson. These are the “pine-apples” and these alone simulate true cones. Their changes will advance rapidly and already their valves may have opened and allowed the aphid larva, which has escaped from its egg at their base, to crawl up and enter. This most remarkable process may be verified by any one who will watch carefully.

Those objects of universal disgust, the Cuckoo-spits, may be shown to have features of interest which will to some extent counteract the repugnance of all juvenile naturalists. Hidden in a mass of iridescent spume there will be found a little greenish insect revelling in the double luxury of warmth and moisture, to which perhaps is added a paradise of many-coloured light. The observer’s attention may be attracted to the fact that the insect has chosen chiefly thistles and nettles as its hosts, these not being likely to be eaten by cattle. Later in the season it will be less careful. The insect here concerned, although a relative of the “plant lice,” is not a true aphid.

June is the month for the Orchis tribe. The Bee must be looked for in its earliest weeks or never. So also of the Spider and the Frog. Others continue in flower much longer, but almost all are in their perfection in June.

Rhododendrons are now plentiful, and the very interesting arrangement by which their anthers open at their points to discharge their pollen may easily be observed. It is characteristic of the whole family of heaths, but as the anthers of rhododendrons are far larger than those of our English heaths, it is best seen in them.

Those who have never observed it before will be amused to be shown the jack-in-the-box manner in which the curled up stamens of the Broom spring out

when the bee touches the shoulders of the petals. June is the month for Broom and both will soon be past.

Amongst our summer visitants the pretty little Turtle-dove is one of the last to arrive. It waits until the season is well settled and rarely comes to us before the middle of May. As perhaps a result of this caution its numbers remain each year much the same. We have now plenty of turtle-doves at Haslemere, whilst all representatives of the Swallow tribe are scarce. Some observers think that Nightingales also have been less frequently heard this spring than usual. Their song, which ceases when the young are hatched and the business of feeding begins, will soon be over for 1906.

Speaking of Birds of Passage, we may say that we shall be glad to receive from any readers in different parts of the kingdom, estimates of the year's abundance of the various species. The spring was remarkably variable, and much of it cold, and this may have had the result of much diminishing the supply of food which would await our guests. So far as our enquiries have gone we believe that there is a general impression that the early spring migrants are this year somewhat defective in number.

Peach and Almond trees, with their leaves curled, distorted and thickened, some yellowish-green, others rosy or purplish, are infested with a fungus, scientifically known as *Exoascus deformans*. It is one of the ascomycetes; in the same genus is included the fungus responsible for the "witches' besoms" of our birch-trees. At maturity the fungus bursts through the cuticle of the leaf, coming to the surface to disperse its spores or seeds. The part of the leaf with the ripe spores upon it, is minutely velvety; the whitish bloom may be easily seen with the unaided eye, but of course individual spores could not be seen without the aid of a powerful microscope.

It is said that this disease is very rarely seen in seasons following an uniformly mild spring.

At the moment of writing we have not observed many oaks defoliated by caterpillars. The following note is taken from the Museum Record Book, June 1, 1899: "Oak-trees badly attacked by the larvæ of the green leaf-roller (*Tortrix viridana*), which cause much damage to the foliage. The continuous falling of the excreta of these small caterpillars sounds like paper being pricked by a fine pin, and is very noticeable in a quiet wood." In 1902 these larvæ were again equally troublesome in this district.

The ingenious work of the leaf-rolling Beetles (*Genera apoderus, Attelabus and Rhynchites*) may be observed at this time of the year. The female, in some species, rolls a leaf into a tube, in others she makes a compact little thimble of the upper half of the leaf, in the centre of which she puts an egg. Hazel, oak and chestnut leaves so folded are not uncommonly seen in this district. Specimens may be usually seen in the vivarium towards the end of the month.

June is a good month for the observation of what are termed Social Flowers. The veronica, in many meadows, especially those in which the grass is kept short by grazing, now exhibits round patches of several yards in diameter, which are beautifully blue. The little mouse ear (*Myosotis*) makes similar patches, but in much less conspicuous tints. These plants appear to have the power not only of spreading themselves, but of excluding intruders, and their territories sometimes show no other form of vegetation. The common daisy is also a social flower, but much less able to keep others at a distance. So also the wild strawberry.

QUESTIONS FOR ANSWERS.

(Continued from p. 31, which see.)

(14) Give the meanings of the following prefixes: (1) sub, (2) pseudo, (3) ob, (4) hypo, (5) hyper, (6) para, (7) ab, (8) aero, (9) con, (10) amphi, (11) ana, (12) pro.

(15) The late Mr. Holyoake tells us that as the result of a street accident in which he was much bruised he remembers “squalling for a fortnight on being taken out of bed.” In another place we read respecting an accident that “a huge dog had loitered behind, and suddenly discovered his master had driven ahead, and he, like a Leming rat, made straight for his master, quite regardless of our being in his way.” Explain the reference to the Leming rat, and rewrite both quotations so as to make them express what you think that the author intended that they should.

(16) What is meant by “a rootless tooth”?

(17) When the number of digits differs on the front and hinder feet of a quadruped, which has usually the most?

(18) Amongst the principal divisions of the mammalian kingdom are Rodentia, Carnivora, Insectivora, Cheiroptera, Primates and Ungulata. Name an English representative of each.

(19) Why are Bees named Anthophila?

(20) Was Captain Cook killed at Owhyhee or in Hawaii?

(21) What do the figures 2123 mean when applied to the teeth, and how many teeth would the animal possess to which that formula would be suitable?

(22) At what time in the morning do Daisies open their flowers?

(23) In the *Times* of September 13, we read, respecting the Sakhalin coast, “The number of walruses and sea-beavers have been greatly reduced by the destructive methods of the American fishers.” What animals are meant by the term “sea-beavers,” and is the name a suitable one?

(24) If you have watched a stableman washing the wheels of a carriage, you will have seen him use an implement for lifting the vehicle from the ground. Why does he use it? What is its name? Explain its mechanism.

(25) If you pour hot water upon a dry sponge it will sink down to half its size, whereas if the water were cold it would swell up. Try the experiment and explain the different results.

(26) When a man's hands are cold he will swing his arms so as to strike the hands violently against the sides of his shoulders. What is this action called, and why is it practised?

NOTICES OF BOOKS RECEIVED.

ORNITHOLOGY.—Messrs. West, Newman and Co., have recently published a very handy and useful “Pocket Book of British Birds,” which we have much pleasure in recommending to field ornithologists. The arrangement followed is that given in Howard Saunders’ well-known “Manual of British Birds.” Species “of which only a few specimens have been observed or obtained in this country” are omitted. The notes are arranged under the heads of localities, haunts, observation, plumage, language, habits, food, nest, site, material, eggs. The size is very convenient for the pocket. Price, 2s. 6d.

THE TRANSACTIONS OF THE BRITISH MYCOLOGICAL SOCIETY for the season 1905 (published, May 19, 1906) contain a full account of the Fungus Foray held at Haslemere during the week ending September 30, 1905.

The specimens collected were exhibited in the Museum. The exhibition was a record one as regards the number of species, as no less than four hundred and eighty-five were identified during the foray, including twenty-five mycetozoa. Four plates, three coloured, accompany the Transactions.

Amongst the species depicted we may mention *Polystictus montagnei*, a new British species found near Haslemere in 1898, and *Sparassis laminosa* (also a new British record) found by Mr. Douglas Taylor on the occasion of the Society’s visit to Woolmer Forest on September 26, last.

Full particulars respecting the Society may be obtained of the Hon. Secretary, Carelton Rea, Esq., B.C.L., M.A., at 34, Foregate Street, Worcester.

DESCRIPTIONS OF MUSEUM SPECIMENS.

[These descriptions are adapted for Museum Labels, and they may be had separately.]

SKULL OF MONTJAC, OR BARKING DEER.

The skull of the little Montjac, or Barking deer, is of interest as showing better than any other the relation of the antler to its pedestal. The pedestal is very long and the antler very small. The latter usually possesses only two tines, the main one and a short stout one which grows near its base. From the front of the pedestal there runs a strong bony ridge down the outer border of the frontal bone as far as the junction with the nasal. This evidently gives strength to the prolonged and rather slender pedestal.

SKULL OF A LLAMA (Camel of America).

The skull of the Llama resembles that of the Camel, and both differ from those of the other ruminants in having incisor teeth in the upper jaw. "These teeth are placed at the side of the intermaxillary bone close to the canines, and agree with them in form" (Van der Hoven, vol. ii., 644).

"There are six incisors only in the lower jaw, and this jaw is undivided."

The camels and llamas form transition species between horses and oxen (ruminants and solid-ungulates).

THE HORNS OF A GNU.

The Gnu's horns are alike in both species and may be known at a glance by their hook-like curves. They pass outwards and downwards and then suddenly curve upwards and forwards. They resemble those of buffaloes and perhaps most closely those of the American bison. They are never very large, and always black. They are of fibrous structure and of large girth at their bases, emulating those of the buffaloes. The Gnu in some of its features resembles a little horse, possessing a mane and having its face, tail, and hindquarters much

like those of a pony. It has, however, a cleft hoof and a beard which, as well as its horns, distinguish it from the horse family.

One of the gnus has a brindled neck and forequarters, pale streaks on a dark ground, and a black and tufted tail; another has a white tail covered with long hair from its base, and shows no brindling. The latter has an almost straight back, whilst the former stands higher in its forequarters like the bison. The horns of the brindled black-tailed gnu do not pass forwards nearly so much as those of the other.

All the Gnus are South African and would appear to bear the same relation to the buffaloes of that continent that the North American bison does to the American buffalo.

They are active but rather awkward animals, and their self-important airs are sometimes amusing.

THE SKULL OF THE DUGONG, OR HALICORE.

This animal is allied to the Manati, both belonging to the order *Sirenia*. They are water-living mammals. The dugong occurs only in Eastern and Australian seas, the manati on the coasts of South America and Africa. The grotesquely misshapen aspect of the skull of the dugong is due to the enormous development of the bones in its upper jaw which carry the cutting teeth (premaxillary bones and incisor teeth), and its clumsy lower jaw. The former bear a tusk in the male, which in the female is present but is never cut. There are no canine teeth, and in the massive lower jaw no incisor teeth are ever cut. The rudiment of one is, however, present in the jaw. It may be noted that the bones carrying the upper incisors do not become united to those of the upper jaw. The back teeth (chewing teeth), are only four, five, or six in number in the dugong, whereas in the manati there may be twenty. A remarkable tendency to vary in their dentition is characteristic of this group of animals, and is no doubt in relation with differences in food. A recently extinct member of the family (Steller's sea-cow) had no teeth at all, but masticated the soft sea-weeds on which it fed by the aid of a horny palate (*Rhytina Stelleri*).

HEAD OF THE WART HOG (*Phacochoerus Æthiopicus*).

The Wart Hog is a native of Africa. His name makes reference to a pair of wart-like excrescences, which are formed, one under each eye. These may be an inch and a half in length.

There is an enormous development of the base of the zygoma. The incisor teeth in the upper jaw are often wanting, and sometimes those in the lower also. The snout is short and square. The so-called warts are fleshy skin-growths and may be large enough to look like ears.

There is another Wart Hog (*Æliani*), met with in Abyssinia. It differs from the Cape Wart Hog in that its incisor teeth in both jaws are more persistent. It has two “warts.”

THE SKULL OF A PIG (*Sus scrofa*).

The skulls of most of the swine family may be recognised by the long face, and the large size, in both jaws, of their dog-teeth or tusks. These are especially large in the male sex, and are often curiously curved, those of the upper jaw upwards, and those of the lower outwards and upwards. The incisor, or biting, teeth vary very much in different species and at different ages. They are often shed early, especially those of the upper jaw. In some pigs the lower incisors are strong and slope directly forwards, as if for digging. The incisor teeth are of less service in the pig than in most animals, and are only exceptionally used for biting or grazing. The pig makes great use of the snout, and the nasal bones are strong and prominent. The molar teeth are well adapted for chewing, and are usually worn flat on their surfaces. Pigs champ but do not ruminate. They are, for the most part, root-eaters. The rim of the orbit is always imperfect. The normal dentition is three incisors, one canine, and seven molars in each jaw. If the upper canine is extracted the lower one will grow into a complete circle and reach the gum close to the root of the tooth. When thus curved it forms an ornament much valued in Fiji.

SKULL OF THE BABIRUSSA.

Note especially that the tusk of the upper jaw grows upward from the first. No part of it is directed into the mouth. This tusk is of extraordinary size, and it curves upwards so as to touch, and sometimes even to pierce, the skull. They are, as compared with those of other swine, slender tusks, more especially the under ones. The upper tusk grows through the skin of the upper lip. In old animals, when it is well curved, it must be useless as a weapon. It may serve to protect the eyes when the animal rushes through brushwood. In the female the tusks are small.

ANSWERS TO CORRESPONDENTS, &c.

CONCHOLOGIST.—Your shells are *Helix nemoralis* (immature), *Hyalinia* (*Vitrea*) *pura*, and *Buliminus obscurus*. *Vitrea pura* is a much smaller shell than *V. nitidula*, the latter may be distinguished from *V. radiatula* by the striæ not being continued from whorl to whorl. *B. obscurus* is much smaller than any member of the genus *Clausilia*. If you examine in May the trunks of beech trees growing on calcareous soils you will find *B. obscurus* and *Clausilia laminata* ascending them in large numbers to spend the summer aloft, coming down again in October to go into hybernation at the base of the trees during the winter months. You will find *C. rugosa* (in modern terminology, *C. bidentata*) equally common on the trees, it is smaller and thinner than *C. laminata*. *B. obscurus* is a short, stout little shell, seldom exceeding 9 millimetres in height.

GARDENER. *Diseases of the Cultivated Chrysanthemum*.—At the present time three diseases are recorded for this country, viz., sclerotium disease (*Sclerotinia sclerotiorum*, Masee); corticium disease (*Peniophora chrysanthemi*, C. B. Plowright), and the only too familiar Rust (*Puccinia hieracii*, Mart). The sclerotium, which first appears as a white mould on the stem just above the ground, forms black lumps within the stem. The stem becomes very brittle and falls. From the black nodules in the following spring, small funnel-shaped brownish fungi with long, weak, dark stems appear. The spores from these settle upon dead organic matter, forming an abundant mycelium which ultimately attacks the base of the stems of chrysanthemums. It is said that fresh stable manure favours the spread of the disease. Diseased stems should be carefully collected and burnt.

The corticium disease also appears on the lower part of the stem, forming a white growth in autumn; Dr. Plowright, who first discovered it, says it resembles a splash of whitewash. It is not confined to the stems but extends into the adjacent soil. Diseased plants are shorter in height and thinner in the stem than healthy; they always die within the year. This disease has hitherto been observed only at King's Lynn in Norfolk, but it is very possible it occurs in other parts of the country. Dr. Plowright remarks that the only treatment is burning the diseased plant. It is useless to separate the apparently healthy shoots.

The well-known chrysanthemum rust first appeared in 1897, and spread with great rapidity in the very dry summer of 1898. The snuff-coloured uredo spores (summer form) are familiar to all cultivators of chrysanthemums. This fungus occurs on many wild plants of the order *Compositæ*, notably the hawk-weeds (*Hieracium*). All diseased plants should be burnt. The disease may be prevented by spraying the new leaves with potassium sulphide solution.

It is to be hoped that the chrysanthemum leaf blight (*Cylindrosporium chrysanthemi*), which has caused so much damage to cultivated plants in Ontario, Canada, will not find its way into this country. It forms large dark patches on the leaves, which turn yellow and hang down; the flower buds do not expand. It is stated that fungicides are useless.

*** END OF THE PROJECT GUTENBERG EBOOK THE HASLEMERE
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